SESSION CHECK

Session 5 — Distances, and how anyone knows

Practical Astronomy Intensive

Name ______________________________ Date ________________

9 questions **15** minutes

They arrived knowing Sirius is 8.6 light years away. This checks whether the session turned that fact back into a measurement — parallax as geometry, the magnitude scale running backwards, and $1/d^2$ quartering rather than halving.

*Circle one letter for each multiple-choice question. Write short answers on the lines. Tasks marked **Practical** are done, not written — your assessor will watch.*

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1 A student writes: "Proxima Centauri is 4.2 light years away, so its light is 4.2 years old." The second half is right. What is a light year?

- a The distance Earth travels along its orbit in one year
- b The speed at which light travels over interstellar distances
- c A length of time — the year light takes to arrive from a star
- d A distance — the 9.46 trillion km light covers in one year

2 You are told 61 Cygni lies 11.4 light years away, measured by parallax. What kind of claim is that?

- a A theory about stellar distances that fits the observations but could be replaced by a better one
- b A figure worked back from the star's colour and spectral type
- c A direct geometric measurement: measure the angular shift, take the 2 AU baseline as the other side of the triangle, and the distance follows
- d An estimate obtained by assuming 61 Cygni is as luminous as the Sun and comparing brightness

3 Voyager 1 is leaving the solar system at about 17 km/s — the fastest thing we have ever launched. Roughly how long would it need to cover the 4.2 light years to Proxima Centauri?

- a About 4.2 years, since that is how long light takes and Voyager is fast
- b About 75,000 years
- c About 400 years, well within a few generations of a crewed mission
- d About 40 years — roughly ten times its journey to the edge of the solar system so far

4. Two identical stars sit in the same field of view, but one is exactly twice as far away as the other. How much fainter does the far one look?

   a. Four times fainter — the same light is spread over four times the area
   b. It looks the same; distance changes the size of a star's disc, not its brightness
   c. Twice as faint — double the distance, half the brightness
   d. Eight times fainter — light spreads through a volume, so $1/d^3$

5. Sirius (apparent magnitude −1.5) looks far brighter than Rigel (apparent magnitude +0.1). Which star actually puts out more energy?

   a. They are equal; magnitude already accounts for distance
   b. Rigel, because its magnitude number is larger and larger numbers mean more power
   c. Rigel, by an enormous margin — Sirius only wins the sky because it is about 8.6 ly away while Rigel is roughly 860 ly
   d. Sirius — it is the brightest star in the night sky, so it must be the most powerful

6. Betelgeuse is distinctly orange-red; Rigel is blue-white. Both are in Orion, both are visible tonight. What does that colour difference tell you?

   a. Betelgeuse is much hotter — red-hot is hotter than blue, as anyone who has seen a stove ring knows
   b. Betelgeuse is closer, and nearer stars look redder
   c. Rigel is much hotter — blue-white is above 10,000 K, red is around 3,000 K
   d. They are made of different elements, and colour reveals composition

7. You measure the same star twice on one clear night: once near the horizon, once near the zenith. The horizon reading is several tenths of a magnitude fainter. What happened?

   a. The night was clear, so extinction is zero; the difference must be instrument drift
   b. The star was farther away when near the horizon, and light falls off with distance
   c. Near the horizon its light crossed several times more atmosphere, which dimmed and reddened it — the star itself did nothing
   d. Nothing physical — a star's brightness is fixed wherever it sits, so one of the two readings must be a mistake

8. A student has a photometer reading for a star and wants its absolute magnitude tonight. What else does she need, and why?

   a. The star's colour, which converts apparent magnitude to absolute magnitude
   b. Nothing — she can subtract 5 from the apparent magnitude, since the scale is linear
   c. The distance — absolute magnitude is a calculation, the brightness the star would show at 10 parsecs, and you cannot get there without knowing how far away it is
   d. Nothing more; absolute magnitude is a direct measurement, just made with a better-calibrated instrument

9. The Greeks argued that if Earth orbited the Sun, the stars would visibly shift over the year — they saw no shift, so Earth must be still. Their observation was honest. Where did the argument fail?

   a. The stars do not shift at all; parallax is an instrumental artefact of modern telescopes
   b. The shift is large but the Greeks had no way to record star positions accurately
   c. The stars are so far away that even the largest stellar parallax is under one arcsecond — far below what an unaided eye can detect
   d. Parallax only appears for stars on the ecliptic, and the Greeks watched the wrong stars